THE LANCET Oncology

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Wilke H, Muro K, Van Cutsem E, et al, for the RAINBOW Study Group. Ramucirumab plus paclitaxel versus placebo plus paclitaxel in patients with previously treated advanced gastric or gastro-oesophageal junction adenocarcinoma (RAINBOW): a double-blind, randomised phase 3 trial. *Lancet Oncol* 2014; published online Sept 18. http://dx.doi.org/10.1016/S1470-2045(14)70420-6.

SUPPLEMENTARY METHODS APPENDIX

Supplement to: Ramucirumab and paclitaxel combination therapy for previously treated advanced gastric or gastro-oesophageal junction adenocarcinoma: The RAINBOW study, a global, double-blind, randomised phase 3 trial

The following methods are excerpts from the study protocol. The original numbering of each section excerpted from the protocol has been retained.

3 STUDY POPULATION SELECTION

3·2 Inclusion Criteria

Each patient must meet the following criteria to be enrolled in this study.

1. The patient has provided signed informed consent and is amenable to compliance with protocol schedules and testing.
2. The patient is at least 18 years of age (or of an acceptable age according to local regulations, whichever is older).
3. The patient has an Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0 or 1 at study entry.
4. The patient has a histopathologically or cytologically confirmed diagnosis of gastric or gastroesophageal junction (GEJ) adenocarcinoma.
5. The patient has metastatic disease or locally advanced, unresectable disease.
6. The patient has experienced documented objective radiographic or clinical disease progression (eg, any new or worsening malignant effusion documented by ultrasound examination) which may be confirmed by pathologic criteria (histology and/or cytology) if appropriate, during first-line therapy, or within 4 months after the last dose of first-line therapy with any platinum/fluoropyrimidine doublet with or without anthracycline (epirubicin or doxorubicin) for unresectable or metastatic disease.
7. The patient has resolution to Grade ≤1 by the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), Version 4·02, of all clinically significant toxic effects of prior locoregional therapy, surgery, or other anticancer therapy.
8. The patient has adequate organ function, defined as:
   - Total bilirubin ≤1·5 times upper limit of normal value (ULN), aspartate transaminase (AST) and alanine transaminase (ALT) ≤3 × ULN for ALT/AST if no liver metastases, ≤5 × ULN if liver metastases;
   - Serum creatinine ≤1·5 ULN or calculated creatinine clearance (per the Cockcroft-Gault formula or equivalent and/or 24-hour urine collection) ≥50 mL/min;
   - Absolute neutrophil count (ANC) $\geq 1{\cdot}5 \times 10^9$/L, hemoglobin ≥ 9 g/dL (5·58 mmol/L; packed red blood cell transfusions are not allowed within one week prior to baseline hematology profile), and platelets $\geq 100 \times 10^9$/L; and
   - International Normalized Ratio (INR) ≤1·5 or Prothrombin time (PT) ≤1·5 × ULN
   - Partial thromboplastin time (PTT/aPTT) ≤ 1.5 × ULN.
9. The patient's urinary protein is ≤1+ on dipstick or routine urinalysis. If urine dipstick or routine analysis indicates proteinuria ≥2+, then a 24-hour urine must be collected and must demonstrate <1000 mg of protein in 24 hours to allow participation in the study.
10. The patient, if female, is surgically sterile, postmenopausal, or compliant with a highly effective contraceptive method (failure rate <1%) during and for 12 weeks after the treatment period (oral hormonal contraception alone is not considered highly effective and must be used in combination with a barrier method). If male, the patient is surgically sterile or compliant with a highly effective contraceptive regimen during and for 6 months after the treatment period. The label (eg, SmPC, USPI, etc) requirements with regard to the methods and duration of contraception during and after treatment with paclitaxel can differ between countries. Country specific requirements will apply only if they are more stringent than those already stipulated in the protocol.

3·3 Exclusion Criteria

Patients who meet any of the following criteria will be excluded from the study.

1. The patient has squamous cell or undifferentiated gastric cancer.
2. The patient has undergone major surgery within 28 days prior to randomization, or central venous access device placement within 7 days prior to randomization.
3. The patient has received any chemotherapy other than platinum and fluoropyrimidine with or without anthracycline for advanced gastric or GEJ adenocarcinoma.
4. The patient has received previous systemic chemotherapy with a cumulative dose of >900 mg/m^2 of epirubicin or >400 mg/m^2 of doxorubicin.
5. The patient has received any previous systemic therapy (including investigational agents) targeting VEGF or the VEGFR signaling pathways. Other previous targeted therapies are permitted, if stopped at least 28 days prior to randomization.
6. The patient has a history of deep vein thrombosis, pulmonary embolism, or any other significant thromboembolism (venous port or catheter thrombosis or superficial venous thrombosis are not considered "significant") during the 3 months prior to randomization.
7. The patient is receiving therapeutic anticoagulation with warfarin, low-molecular weight heparin or similar agents. Patients receiving prophylactic, low-dose anticoagulation therapy are eligible provided that the coagulation parameters defined in the inclusion criteria (INR $\leq 1{\cdot}5$ and PTT/aPTT $\leq 1{\cdot}5 \times$ ULN) or (PT $\leq 1{\cdot}5 \times$ ULN and PTT/aPTT $\leq 1{\cdot}5 \times$ ULN) are met.
8. The patient is receiving chronic therapy with nonsteroidal anti-inflammatory agents (NSAIDs, eg, indomethacin, ibuprofen, naproxen or similar agents) or other anti-platelet agents (eg, clopidogrel, ticlopidine, dipyridamole, anagrelide). Aspirin use at doses up to 325 mg/day is permitted.
9. The patient has significant bleeding disorders, vasculitis, or had a significant bleeding episode from the gastrointestinal tract within 3 months prior to study entry.
10. History of gastrointestinal perforation and/or fistulae within 6 months prior to randomization.
11. The patient has symptomatic congestive heart failure (New York Heart Association II-IV) or symptomatic or poorly controlled cardiac arrhythmia.
12. The patient has experienced any arterial thrombotic event, including myocardial infarction, unstable angina, cerebrovascular accident, or transient ischemic attack, within 6 months prior to randomization.
13. The patient has uncontrolled arterial hypertension $\geq$150/$\geq$90 mm Hg despite standard medical management.
14. The patient has a serious or non healing wound or peptic ulcer or bone fracture within 28 days prior to randomization.
15. The patient has a bowel obstruction, history or presence of inflammatory enteropathy or extensive intestinal resection (hemicolectomy or extensive small intestine resection with chronic diarrhea), Crohn's disease, ulcerative colitis, or chronic diarrhea.
16. The patient has a serious illness or medical condition(s) including, but not limited to the following:
    - Known human immunodeficiency virus infection or acquired immunodeficiency syndrome-related illness
    - Active or uncontrolled clinically serious infection
    - Previous or concurrent malignancy except for basal or squamous cell skin cancer and/or in situ carcinoma of the cervix, or other solid tumors treated curatively and without evidence of recurrence for at least 3 years prior to the study
    - Uncontrolled metabolic disorders or other nonmalignant organ or systemic diseases or secondary effects of cancer that induce a high medical risk and/or make assessment of survival uncertain
    - Other severe acute or chronic medical or psychiatric condition or laboratory abnormality that may increase the risk associated with study participation or study drug administration, or may interfere with

the interpretation of study results, and in the judgment of the Investigator would make the patient ineligible for entry into this study

- History or evidence of known central nervous system metastases or carcinomatous meningitis;
- Known allergy or hypersensitivity to monoclonal antibody treatment or any components used in the ramucirumab DP preparation
- Known allergy or hypersensitivity to paclitaxel or any components used in the paclitaxel preparation or other contraindication for taxane therapy.

17. The patient is pregnant or breastfeeding.
18. The patient is currently enrolled in, or discontinued within the last 28 days from a clinical trial involving an investigational product or non-approved use of a drug, or concurrently enrolled in any other type of medical research judged not to be scientifically or medically compatible with this study. Patients participating in surveys or observational studies are eligible to participate in this study.

10 DOSE MODIFICATIONS

10·1 Treatment Requirements and Delays

10·1·1 Requirements Prior to Day 1 of Each Treatment Cycle

Prior to Day 1 of each administration of study therapy, hematology, urine protein (dipstick), liver, and renal function must be adequate (see Table and Table *1*), and all toxicities (except for alopecia, and venous thrombotic events, hypertension, proteinuria [please see Section **Error! Reference source not found.** for detailed instructions on the management of these AEs]) must have resolved to Grade <2 or baseline. Preinfusion laboratory data may not be older than 36 hours.

Table 5: **Criteria for paclitaxel treatment on Day 1 of each cycle**

Neutrophils:	$\geq 1{\cdot}5 \times 10^9$ / L
Platelets:	$\geq 100 \times 10^9$ / L
Serum Creatinine:	$\leq 1{\cdot}5 \times$ ULN or calculated creatinine clearance $\geq$50 ml/min
Bilirubin:	$\leq 1{\cdot}5 \times$ ULN
AST/ALT:	$\leq 3 \times$ ULN for ALT/AST if no liver metastases, $\leq 5 \times$ ULN if liver metastases
Paclitaxel-related Toxicities/AEs:	NCI-CTCAE Version 4·02 Grade <2 or baseline (except for alopecia.) Anemia $\leq$Grade 2

Table 1:**Criteria for ramucirumab DP treatment on Day 1 of each cycle**

Urine protein:	Dipstick <2+ or protein level <2 g/24 h (see Section **Error! Reference source not found.**)
Ramucirumab DP-related toxicities/AEs:	NCI-CTCAE Version 4·02 Grade < 2 or baseline (except for hypertension, venous thromboembolic events, proteinuria, as detailed in Section **Error! Reference source not found.**).

If one or more criteria outlined in the tables above are not met at the time of a planned treatment, the following general rules for the management of treatment delays apply:

- In the case of paclitaxel-related toxicity or abnormal laboratory values (see Table), blood counts and serum chemistry will be repeated weekly. The start of the next cycle will be delayed until recovery to the values stated above. However, ramucirumab DP/placebo therapy should continue as scheduled until the next cycle has resumed. When the subsequent cycle of paclitaxel is initiated, administration of ramucirumab DP/placebo and paclitaxel will be resynchronized according to the study design described in this protocol (ie, the cycle will begin at Day 1 for both ramucirumab DP and paclitaxel, even if this requires ramucirumab DP to be administered on consecutive weeks). In case of discontinuation of paclitaxel for any reason, a new cycle will be started on Day 29 (Day 1 of the new cycle) with the administration of ramucirumab DP monotherapy, provided the criteria outlined in Table 1 are met.

- In the case of ramucirumab DP/placebo-related toxicity, ramucirumab DP/placebo will be delayed for one week and administered on Day 8 (and Day 15) of the treatment cycle provided that ramucirumab DP/placebo-related toxicities have resolved to Grade <2 or baseline (except for hypertension, venous thromboembolic events, and proteinuria, as detailed in Section **Error! Reference source not found.**). If toxicities have not resolved on Day 8, ramucirumab DP/placebo will be delayed for another week and administered on Day 15 according to the schedule described in this protocol. In both cases, paclitaxel will continue according to the planned schedule.

- If a patient cannot be treated with one component of the study therapy (paclitaxel or ramucirumab DP) for more than 28 days from the last administered dose, that component will be permanently discontinued. The other agent should be continued, with the patient remaining on study, if clinically indicated.

- If the start of a cycle is delayed due to toxicity, radiographic assessment of disease response should <u>not</u> be delayed, but should be performed every 6 weeks (±3 days) following the first dose of study therapy for the first 6 months following the first dose, and every 9 weeks (±3 days) following the first dose thereafter, until documentation of PD.

10·1·2 Requirements Prior to Days 8 and 15 of Each Treatment Cycle

Prior to Day 8 and 15 of each administration of study therapy, hematology, urine protein (dipstick), and liver function must be adequate (see Table 2 and

Table 3), and all toxicities (except for alopecia, and venous thrombotic events, hypertension, and proteinuria [please see Section **Error! Reference source not found.**] for detailed instructions on the management of these AEs]) must have resolved to Grade <2 or baseline. Preinfusion laboratory data may not be older than 36 hours.

Table 2: **Criteria for paclitaxel treatment on Days 8 and 15 of each cycle**

Parameter	Criterion
Absolute neutrophil count	$\geq 1{\cdot}0 \times 10^9$/L
Platelet count	$\geq 75 \times 10^9$/L
Bilirubin	$\leq 1{\cdot}5 \times$ ULN
AST/ALT	$\leq 3 \times$ ULN, or $< 5 \times$ ULN if the aminotransferase elevation is due to liver metastases
Paclitaxel-related toxicities/AEs	NCI-CTCAE Version 4·02 Grade <2 or baseline (except for alopecia). Anemia ≤ Grade 2

Table 3: **Criteria for ramucirumab DP treatment on Day 15 of each cycle**

Parameter	Criterion
Urine protein	Dipstick <2+ or protein level <2 g/24 h (see Section **Error! Reference source not found.**)
Ramucirumab DP-related toxicities/AEs	NCI-CTCAE v 4·02 Grade <2 or baseline (except for hypertension, venous thromboembolic events, proteinuria, as detailed in Section **Error! Reference source not found.**).

If one or more criteria outlined in the tables above are not met at the time of a planned treatment, the following general rules for the management of treatment delays apply:

- In the case of paclitaxel-related toxicity or abnormal laboratory values (Table 2) on Day 8 or 15, paclitaxel will be skipped at that day. No dose reductions are allowed within a given cycle.
- In the case of ramucirumab DP-related toxicity on Day 15 (
  Table 3), paclitaxel will be administered according to the planned schedule but ramucirumab DP will be delayed for one week and administered on Day 22 of the treatment cycle, provided that ramucirumab DP-related toxicities have resolved to Grade <2 or baseline (except for hypertension, venous thromboembolic events, or proteinuria, as detailed in Section **Error! Reference source not found.**). If toxicities have not resolved on Day 22, ramucirumab DP will be skipped for that cycle and administered on Day 1 of the following cycle provided that ramucirumab DP-related toxicities have resolved to Grade <2 or baseline.

10·2 General Dose Modifications

10·2·1 Ramucirumab DP

Dose modifications are permitted for ramucirumab DP/placebo in the setting of non-life-threatening and reversible Grade 3 clinical AEs (eg, fever) considered to be at least possibly related to ramucirumab DP/placebo and that resolve to Grade ≤1 or pretreatment baseline within one treatment cycle (approximately 28 days). If a Grade 4 AE occurs and is deemed at least possibly related to ramucirumab DP/placebo, then ramucirumab DP/placebo should be discontinued except in the specific case of Grade 4 fever or Grade 4 laboratory abnormalities. If Grade 4 fever or laboratory abnormalities resolve to Grade ≤1 or pretreatment baseline within 1 treatment cycle (approximately 28 days), treatment with investigational product may be continued at the discretion of the investigator. In these settings, ramucirumab DP/placebo may be readministered. If a second instance of such an event occurs, ramucirumab DP/placebo should be subsequently readministered at a dose of 6 mg/kg every other week. A second dose reduction to 5 mg/kg every other week is permitted for this level of event (Grade 3 or 4 event). If the dose of ramucirumab DP/placebo is reduced because of potentially related AEs, subsequent dose increases are not permitted. Note that this section of the protocol pertains only to general dose modifications. Criteria for dose reduction in the setting of various specific AEs may be found in Section **Error! Reference source not found.**. Ramucirumab DP/placebo will be permanently discontinued in case of any ramucirumab DP/placebo-related event that is deemed life-threatening, regardless of grade.

Patients who enter the study with symptoms or laboratory values equivalent to NCI-CTCAE Version 4·02 Grade 1 or 2 AEs should not have dose reductions related to the persistence or mild worsening of those symptoms or laboratory values; dose reductions may be warranted if worsening of symptoms or laboratory values is clinically significant in the opinion of the investigator. Except in the case of proteinuria, as described in Section **Error! Reference source not found.**, asymptomatic laboratory abnormalities should not result in dose interruptions, modifications, or discontinuation of study therapy unless determined by the investigator to be clinically significant or life-threatening.

10·2·2 Paclitaxel

No dose modification for paclitaxel is allowed within a given cycle. The paclitaxel dose will be reduced by 10 mg/m^2 for the following cycle when NCI-CTCAE (Version 4·02) Grade 4 hematological toxicity or Grade 3 paclitaxel-related nonhematological toxicity (except for alopecia) is observed. If the dose of paclitaxel is reduced because of potentially related AEs, subsequent dose increases are not permitted. Paclitaxel will be permanently discontinued if dose reduction to less than 60 mg/m^2 would be required, or in case of any paclitaxel-related event that is deemed life-threatening, regardless of grade.

10·3 Removal of Patients from Therapy or Assessment

10·3·1 Removal from All Study-Related Therapy

The investigator should withdraw a patient from all study-related therapy for any of the following reasons:

- Radiographic progression as defined in Section **Error! Reference source not found.** or symptomatic progression of disease;
- The Sponsor or investigator terminates the study;
- An unacceptable AE/toxicity (eg, a persistent moderate toxicity that is intolerable to the patient), which in the opinion of the investigator cannot be attributed to a specific study agent;
- A Grade 3-4 infusion related reaction which in the opinion of the investigator cannot be attributed to a single study agent (considered at least possibly related to each study agent);
- Any therapy-related event that is deemed life-threatening, regardless of grade;
- Any event which would require study therapy to be modified by more than two dose reductions or to be held for more than 28 days from the last administered dose, and is considered at least possibly related to both study agents (paclitaxel and ramucirumab DP/placebo);
- An intercurrent illness or change in the patient's condition that renders the patient unsuitable for further treatment in the opinion of the investigator;
- Protocol noncompliance with this study protocol. Patients who miss appointments shall be contacted by site personnel to determine the reason for the missed appointment and to try to reschedule the appointment. The date(s) the patient was contacted and the type of contact used should be recorded in the study documentation; and/or
- Withdrawal of consent (if the patient withdraws consent to treatment, he or she may still enter survival follow-up if follow-up consent is not withdrawn).

10·3·2 Removal From Ramucirumab DP/Placebo Therapy

The investigator may withdraw a patient from ramucirumab DP/placebo therapy for any of the following reasons:

- An unacceptable AE/toxicity (eg, a persistent moderate toxicity that is intolerable to the patient) that is, in the opinion of the investigator, clearly attributed to ramucirumab DP/placebo (and considered unrelated or unlikely related to paclitaxel);
- A Grade 3-4 infusion reaction that is clearly attributed to ramucirumab DP (and unrelated or unlikely related to paclitaxel);
- A Grade 3-4 arterial thrombotic event;
- Any Grade 4 venous thrombotic event or a Grade 3 venous thrombotic event that is considered by the investigator to be life-threatening, or symptomatic and not adequately treated by anticoagulation therapy;
- Any venous thrombotic event occurring or intensifying during anticoagulant therapy
- A Grade 3-4 bleeding or hemorrhagic event;

- Grade 4 hypertension or persistent/recurrent hypertension as detailed in Section **Error! Reference source not found.**;
- Proteinuria >3 g/24 h or persistent/recurrent proteinuria >2 g/24 h, as detailed in Section **Error! Reference source not found.**;
- Any Grade 4 (life-threatening) non-hematologic toxicity considered by the investigator to be possibly, probably or definitely related to ramucirumab DP/placebo;
- Gastrointestinal perforation;
- Any Grade 3-4 events consistent with congestive heart failure (CHF);
- Any event which would warrant ramucirumab DP/placebo therapy to be modified by more than two dose reductions or to be held for more than 28 days from the last administered dose, and is clearly attributed to ramucirumab DP/placebo (ie, recurrent or persistent hypertension or proteinuria); and/or
- New occurrence of hepatic encephalopathy and/or hepatorenal syndrome resulting from liver cirrhosis.

10·3·3 Removal from Paclitaxel Therapy

The investigator may withdraw a patient from paclitaxel therapy for any of the following reasons:

- An unacceptable AE/toxicity (eg, a persistent moderate toxicity that is intolerable to the patient) and is, in the opinion of the investigator, clearly attributed to paclitaxel (and considered unrelated or unlikely related to ramucirumab DP/placebo);
- A Grade 3-4 infusion reaction that is clearly attributed to paclitaxel (and considered unrelated or unlikely related to ramucirumab DP/placebo); and/or
- Any event which would warrant paclitaxel therapy to be modified by more than two dose reductions or to be held for more than 28 days from the last administered dose, and is clearly attributed to paclitaxel (ie, recurrent or persistent neuropathy).

The ImClone Medical Monitor will be consulted before a patient is discontinued from the study due to paclitaxel intolerance.

10·3·4 Continuation of Therapy On-Study in the Setting of Discontinuation of One Component of Therapy

If one therapeutic agent is permanently discontinued secondary to toxicity, then therapy with the other study agents should continue and the patient should remain on-study with full adherence to all protocol-related requirements. Patients who have paclitaxel discontinued may continue to receive ramucirumab DP/placebo on-study, provided they continue to meet entry criteria and do not have evidence of disease progression. Patients who have ramucirumab DP/placebo discontinued (ie, in the setting of Grade 4 hypertension or proteinuria) may receive ongoing paclitaxel therapy on-study, provided they continue to meet entry criteria and do not have evidence of disease progression.

After termination of study therapy, the patient will be treated as clinically indicated by the investigator or referring physician. All patients will be followed until resolution or stabilization of any SAE or study-related toxicity. If the reason for withdrawal from the trial is the death of the patient, the two options for categorizing withdrawal are either progressive disease or an AE (more than one AE may be documented as a reason for withdrawal). Only one event will be recognized as the cause of death. If a patient is discontinued from all study therapy:

- The reason(s) for discontinuation should be documented in the patient's medical record and eCRF;
- An end-of-therapy evaluation should be performed as described in Section **Error! Reference source not found.**;
- For patients who discontinue for reasons other than progression of disease, radiographic assessment for tumor response should continue every 6 weeks following the first dose of study therapy (±3 days) for the

first 6 months following the first dose, and every 9 weeks (±3 days) thereafter, until there is radiographic documentation of progressive disease (PD);

- A follow-up evaluation should be performed at least 30 days after the last dose of study therapy as described in Section **Error! Reference source not found.**; and

  The patient should be followed for survival at regularly scheduled intervals (at least every 8 weeks [+ 0-7 days]) for as long as he or she remains alive (until study completion as defined in Section **Error! Reference source not found.**); for additional details, please see Section **Error! Reference source not found.**.

6 STUDY PROCEDURES

6·4 Physical Examination

Complete Physical Examination – evaluation by body system, height (screening/baseline only), weight, and body surface area.

ECOG Performance Status – according to the ECOG PS criteria.

Toxicity/AE Assessment – including NCI-CTCAE Version 4·02 grade.

6·5 Vital Signs

Vital Sign Measurements – include temperature, pulse rate, respiration rate, and blood pressure.

6·6 Clinical Tests

6·6·1 Laboratory Parameters

Hematology Profile – includes a complete blood count with differential, white blood cell count, and platelet count.

Coagulation Profile – includes INR or PT (whichever is selected should be followed throughout), and PTT.

Full Serum Chemistry Profile – includes sodium, potassium, blood urea nitrogen, phosphorous, creatinine, glucose, total protein, albumin, uric acid, AST, ALT, alkaline phosphatase, total bilirubin, lactate dehydrogenase, calcium, and magnesium.

Urinalysis - routine dipstick measurements and, if clinically indicated, microscopic analysis. If urine dipstick or routine analysis indicates proteinuria ≥2+, a 24-hour urine collection (to assess protein) must be obtained.

Pregnancy Test – serum or urine pregnancy test, minimum sensitivity 25 IU/L or equivalent units of β-human chorionic gonadotropin (β-HCG), for women of childbearing potential (WOCBP).

6·6·2 Other Tests

Electrocardiogram (ECG) – to be obtained at screening/baseline (not more than 21 days prior to randomization), prior to the first infusion of Cycle 2 and 3, every 8 weeks afterwards, at the discontinuation of chemotherapy, and at the discontinuation of ramucirumab DP/placebo.

Echocardiogram or Multiple Gated Acquisition Scan (MUGA) – to be obtained at screening/baseline (not more than 21 days prior to randomization), prior to the first infusion of Cycle 3 only, at the discontinuation of chemotherapy, and at the discontinuation of ramucirumab DP/placebo.

Table 2: **Study events table – study period**

Study Procedures	Screening/Baseline (Prior to Randomization)				Within 7 Days Prior to Day 1	Cycles (Repeat every 28 days [4 weeks])				Every 6 Weeks	Every 8 Weeks	End of Therapy	30-Day Safety Follow-Up[a]	End of Study[b]	Survival Follow-Up at least every 8 Weeks[t]
	Within 28 Days	Within 21 Days	Within 14 Days	Within 7 Days		Wk. 1 (D 1)	Wk. 2 (D 8)	Wk. 3 (D 15)	Wk. 4 (D 22)	(±3 d)	(±3 d)	(+0-7 d)	(+0-7 d)	(+0-7 d)	(+0-7 d)
Clinical Evaluations															
Informed Consent	X[c]				Randomization										
Demography	X														
Medical History	X[d]														
Electrocardiogram		X				X[e]					X[e]	X[e]			
Echocardiogram/MUGA		X[f]									X[f]	X[f]			
ECOG Performance Status		X				X						X	X		
Concomitant Medication Assessment			X[g]			X	X	X				X	X		
Physical Exam, Height & Weight			X[h]			X[h]						X[h]	X[h]		
Vital Signs			X			X[i]	X[i]	X[i]				X	X		
Toxicity Assessments/AEs			X[d]			X	X	X				X	X	X	X[j]
PRO Assessment			X[k]							X[k,l]		X[k]			
Laboratory Evaluations															
Hematology Profile				X	Randomization	X[m]	X[m]	X[m]				X	X		
Coagulation Profile				X							X	X			

ALT, AST, Total Bilirubin		X		X[n]	X[n]	X[n]				X	X		
Full Serum Chemistry Profile		X		X						X	X		
Urinalysis	X[o]			X[o]		X[o]				X[o]	X[o]		
Pregnancy Test		X[p]							X[p]		X		
PK/Immunogenicity/PD Sample	Please see Section **Error! Reference source not found.**, **Error! Reference source not found.**												

Study Procedures	Screening/Baseline (Prior to Randomization)				Within 7 Days Prior to Day 1	Cycles (Repeat every 28 days [4 weeks])				Every 6 Weeks (±3 d)	Every 8 Weeks (±3 d)	End of Therapy (+0-7 d)	30-Day Safety Follow-Up[a] (+0-7 d)	End of Study[b] (+0-7 d)	Survival Follow-Up at least every 8 Weeks[t] (+0-7 d)
	Within 28 Days	Within 21 Days	Within 14 Days	Within 7 Days		Wk. 1 (D 1)	Wk. 2 (D 8)	Wk. 3 (D 15)	Wk. 4 (D 22)						
Efficacy Assessments															
Survival Information					Randomization										X
Imaging / Tumor Assessments		X								X[l]				X[q]	
Treatment Administration															
Administer Ramucirumab DP/Placebo						X		X							
Administer Paclitaxel						X[r,s]	X	X[r,s]							

a. To be performed a minimum of 30 days and a maximum of 37 days after the last dose of study therapy.

b. The end-of-study visit should occur within 7 days after the documentation of progressive disease if progressive disease occurs subsequent to the 30-day safety follow-up visit. For patients who discontinue treatment due to documented progressive disease, or who discontinue treatment for other reasons but experience documented progressive disease within 30 days after the last dose of study therapy, the end-of-study visit may be concurrent with the 30-day safety follow-up visit.

c. Written informed consent must be obtained prior to any study-related procedures or evaluations. ICF signature alone does not define start of screening period, but first study-related procedure or evaluation date is to be used for IV/WRS call/date of screening visit.

d. Any pre-existing toxicity within 28 days prior to randomization should be documented and recorded as part of the medical history at baseline.

e. Additional ECGs will be collected prior to the infusion of Cycle 2 and 3 and every 8 weeks thereafter, at the discontinuation of chemotherapy, and at the discontinuation of ramucirumab DP/placebo.

f. To be performed at screening/baseline (within 21 days prior to randomization), prior to the first infusion of Cycle 3 only, and at the discontinuation of chemotherapy and at the discontinuation of ramucirumab DP/placebo.

g. Including all medications taken within 28 days prior to randomization.

h. Height measurements to be performed at screening/baseline only.

i. To be obtained at every treatment visit, before and at the completion of each infusion of ramucirumab DP/placebo and before and at the completion of chemotherapy administration.

j. All serious adverse events and all adverse events considered at least possibly related to any study therapy will be followed until resolution, stabilization, return to baseline, or until deemed irreversible.

k. To be completed at the beginning of the visit, before blood sampling or any extensive contact and consultation with the clinician/study investigator; such encounters may thereafter bias patient responses. The instruments will be administered together and in sequence order, with the EORTC QLQ-C30 presented first, followed by presentation of the EQ-5D.

l. Imaging studies, tumor assessments, and PRO evaluations should be performed as scheduled every 6 weeks (± 3 days) following first dose of study therapy until radiographically documented progression of disease, even if therapy is delayed. Imaging should be performed every 6 weeks (± 3 days) only for the first 6 months following the first dose, and every 9 weeks (± 3 days) thereafter. After 6 months, PRO evaluations can be performed approximately every 6 to 9 weeks.

m. Hematology profile to be collected within 36 hours (blood may be collected the morning of study therapy or the day before, results to be available and evaluated prior to start of infusion) prior to treatment on Days 1, 8, and 15 of each cycle until discontinuation of paclitaxel and prior to the start (Day 1) of each 28-day cycle only thereafter (Arms A and B).

n. ALT, AST, and total bilirubin to be collected within 36 hours (blood may be collected the morning of study therapy or the day before, results to be available and evaluated prior to start of infusion) prior to treatment on Days 1, 8, and 15 of each cycle until discontinuation of paclitaxel, prior to the start (Day 1) of each 28-day cycle thereafter (with full serum chemistry profile), and at end of therapy (EoT).

o. Routine dipstick measurements and, if clinically indicated, microscopic analysis. If urine dipstick or routine analysis indicates proteinuria ≥ 2+, a 24 hour urine collection (to assess protein) must be obtained.

p. Or according to institutional guidelines (serum pregnancy test for WOCBP mandatory at baseline).

q. For patients removed from the study due to symptomatic deterioration characterized as progressive disease in the absence of radiographic documentation of progressive disease.

r. Paclitaxel must not be started earlier than 60 minutes after completion of the ramucirumab DP/placebo infusion.

s. The 1-hour observation period between administration of ramucirumab DP/placebo and paclitaxel is mandatory for the first two cycles, but may be omitted in Cycle 3 and beyond provided there has been no evidence of infusion-related reaction.

Determination of Ramucirumab Antidrug Antibodies

The formation of antidrug antibodies (ADAs) was assessed in validated immunogenicity assays. A four-tier testing scheme was utilized. In tier 1, the ADA screening assay used a modified affinity capture elution format. First, serum samples were incubated with human VEGF-110, a stable recombinant version of the endogenous VEGF, to minimize assay interference from endogenous VEGF receptor 2. Following this step, serum samples potentially containing ADAs and ramucirumab were acidified to dissociate immune complexes. The acidified sample was then neutralized on an ELISA plate coated with ramucirumab, allowing the ADAs to bind. After washing free drug away, bound ADAs were eluted off the plate by acid treatment. Eluted ADAs were then neutralized on a second ELISA plate where they were allowed to bind to the plate surface. These plate-bound ADAs were detected by addition and binding of biotin-labeled ramucirumab followed by development with horseradish peroxidase-labeled streptavidin and enzyme substrate. Validation experiments for the tier 1 screening assay demonstrated a sensitivity of 3·91 ng/mL and a drug tolerance of 250 µg/mL of ramucirumab in the presence of 500 ng/mL of affinity-purified ADA from hyperimmunized monkey serum.

The tier 1 screening assay format was also used in tier 2 testing, with a slight modification involving addition of excess nonlabeled ramucirumab at the biotinylated ramucirumab step to confirm the specificity of presumed ADA identified during tier 1 screening. The tier 1 screening format was also utilized in tier 3 testing by measuring the titer of specific ADA confirmed in tier 2. The titer was reported as the highest dilution of the sample that still yielded a response greater than the cut point of the assay.

Samples that were confirmed as being positive for ADA, tier 1 and tier 2 positive, were further evaluated in the neutralizing ADA assay (tier 4). The tier 4 assay format relies on direct competition between the binding of the ruthenium-labeled VEGF receptor 2 (therapeutic target) and binding of ADA at the active site of ramucirumab. In the absence of ADA, the ruthenium-labeled VEGF receptor 2 binds with ramucirumab to produce an electrochemiluminescent signal. However, in the presence of neutralizing ADA, the ADA competes with the ruthenium labeled VEGF receptor 2 for binding at the active site of ramucirumab and the overall signal is diminished. Validation experiments for the tier 4 neutralizing assay demonstrated a sensitivity of 106 ng/mL and a drug tolerance of 0·078 µg/mL for ramucirumab in the presence of 500 ng/mL of affinity-purified ADA from hyperimmunized monkey serum. Addition of endogenous target was shown to have no effect in the presence of 500 ng/mL affinity-purified ADA from hyperimmunized monkey serum.

SUPPLEMENTARY RESULTS APPENDIX

Supplement to: Ramucirumab and paclitaxel combination therapy for previously treated advanced gastric or gastro-oesophageal junction adenocarcinoma: The RAINBOW study, a global, double-blind, randomised phase 3 trial

Table 1S*: Reasons for treatment discontinuation

	Ramucirumab plus Paclitaxel (N=330)	**Placebo plus Paclitaxel (N=335)**
Progressive disease	236 (72%)	255 (76%)
Death	12 (4%)	13 (4%)
Adverse event	39 (12%)	38 (11%)
Withdrawal of consent	23 (7%)	13 (4%)
Other	3 (<1%)	3 (<1%)
Lost to follow-up	0	1 (<1%)

Data are n (%), unless otherwise indicated. *At the data cutoff date of July 12, 2013, 313 (95%) patients in the ramucirumab plus paclitaxel group discontinued therapy, and 323 (96%) patients in the placebo plus paclitaxel group discontinued therapy.

Table 2S*: Multivariate analysis of overall survival adjusting for significant covariates

Variable	p value	Hazard ratio (95% CI)
Study treatment (ramucirumab plus paclitaxel *vs* placebo plus paclitaxel)	0·0010	0·745 (0·626–0·888)
Geographic region†	<0·0001	
Region 2 *vs* Region 1		0·985 (0·683–1·422)
Region 3 *vs* Region 1		0·604 (0·494–0·740)
ECOG performance status (0 *vs* 1)	<0·0001	1·448 (1·205–1·740)
Weight loss prior 3 months (≥10% *vs* <10%)	0·0025	1·452 (1·411–1·848)
Number of metastatic sites (≤2 *vs* ≥3)	0·0114	0·783 (0·648–0·946)
Presence of ascites (yes *vs* no)	<0·0001	1·502 (1·237–1·824)
Tumour differentiation	0·0285	
Well *vs* poorly		0·713 (0·495–1·026)
Moderately *vs* poorly		0·755 (0·615–0·926)
Unknown *vs* poorly		0·841 (0·580–1·218)
Prior gastrectomy (yes *vs* no)	0·0036	0·758 (0·629–0·913)

*The covariates in the model were selected from potential variables [geographical region, disease measurability, time to progression on first-line therapy, gender, age, race, ECOG performance status, weight loss, primary tumor location, prior first-line chemotherapy (doublets/triplets), histologic subtype, number of metastatic sites, peritoneal metastasis, liver metastasis, presence of ascites, tumor differentiation, number of previous treatment lines, prior gastrectomy] using stepwise Cox regression with entry p value 0·05 and exit p value 0·1, and the treatment variable was added into the final model after selection of covariates. †Region 1: Europe, Israel, Australia, and United States; Region 2: Argentina, Brazil, Chile, and Mexico; Region 3: Japan, South Korea, Hong Kong, Singapore, and Taiwan.

Table 3S*: Cox regression analysis of overall survival adjusting for significant factors

	Adjusted for individual factors		Adjusted for all previous factors	
	Hazard ratio	**p value**	**Hazard ratio**	**p value**
Unadjusted, unstratified	-	-	0.822	0.0258
ECOG performance status*	0.785	0.0062	0.785	0.0062
Region*	0.797	0.0104	0.768	0.0029
Ascites*	0.798	0.0107	0.746	0.0010
Number of metastatic sites	0.805	0.0143	0.743	0.0008
Tumour differentiation	0.812	0.0185	0.742	0.0008
Weight loss	0.814	0.0198	0.742	0.0008
Prior gastrectomy	0.836	0.0422	0.745	0.0010

*ECOG performance status, region, and presence of ascites were the main contributors to the difference between adjusted and unadjusted efficacy estimates.

Table 4S*: Baseline ECOG performance status and presence of ascites by treatment group

		Ramucirumab plus Paclitaxel N=330 n (%)	**Placebo plus Paclitaxel N=335 n (%)**
ECOG performance status	0	117 (35.5)	144 (43.0)
	1	213 (64.5)	191 (57.0)
Presence of ascites	Yes	130 (39.4)	107 (31.9)

*There was a slight imbalance in ECOG performance status and ascites favouring the placebo plus paclitaxel group.

Table 5S*: Multivariate analysis of progression-free survival adjusting for significant covariates

Variable	p value	Hazard ratio (95% CI)
Study treatment (ramucirumab plus paclitaxel *vs* placebo plus paclitaxel)	<0·0001	0·599 (0·506–0·708)
Gender (male *vs* female)	0·0106	0·790 (0·659–0·947)
Weight loss prior 3 months (≥10% *vs* <10%)	0·0241	1·299 (1·305–1·631)
Number of metastatic sites (≤2 *vs* ≥3)	0·0145	0·800 (0·668–0·957)
Liver metastasis (yes *vs* no)	0·0020	1·317 (1·106–1·569)

*The covariates in the model were selected from potential variables as pre-specified in the statistical analysis plan, using stepwise Cox regression method with entry p value 0·05 and exit p value 0·1, and the treatment variable was added into the final model after selection of covariates.

***Table 6S*: Summary of study drug exposure**

	Ramucirumab plus paclitaxel (n=327)		Paclitaxel plus placebo (n=329)	
	Ramucirumab*	**Paclitaxel†**	**Placebo***	**Paclitaxel†**
Duration of therapy, weeks				
Mean (SD)	23·0 (18·5)	21·7 (16·7)	16·4 (13·8)	16·5 (14·4)
Median	18·0	17·7	12·0	12·0
Range	2–102	2–94	2–101	2–103
Number of infusions received				
Mean (SD)	11·1 (8·8)	13·7 (10·63)	8·1 (6·8)	11·3 (9·9)
Median	9·0	11·0	6·0	9·0
Range	1–53	1–55	1–46	1–67
Cumulative dose				
Mean (SD)	87·8 (68·5)	1067·9 (815 2)	64·3 (54·0)	898·0 (787·4)
Median	70·0 (mg/kg)*	813·0 (mg/m^2)†	49·0 (mg/kg)*	714·0 (mg/m^2)†
Range	8–420	79–4399	8–363	64–5330
Relative dose intensity, %				
Mean (SD)	97 (8)	84 (15)	98 (6)	90 (12)
Median	99*	88†	100*	93†
Range	64–114	45–104	59–112	38–107

*Based on last available weight prior to each infusion. †Based on last available body surface area prior to each infusion. Dose intensity and cumulative dose calculations for placebo were based on the volume of placebo administered. SD=standard deviation.

***Table 7S*: Post-discontinuation systemic antineoplastic therapy**

	Ramucirumab plus Paclitaxel	Placebo plus Paclitaxel
All patients	(N=330)	(N=335)
Any treatment	158 (48%)	154 (46%)
Biologic	23 (7%)	18 (5%)
Chemotherapy†	158 (48%)	152 (45%)
Irinotecan	124 (38%)	117 (35%)
Fluorouracil	83 (25%)	74 (22%)
Other chemotherapies‡	125 (38%)	113 (34%)
Other	1 (<1%)	7 (2%)
Regions 1 + 2*	(n=221)	(n=221)
Any treatment	83 (38%)	79 (36%)
Biologic	9 (4%)	6 (3%)
Chemotherapy†	83 (38%)	78 (35%)
Irinotecan	63 (29%)	55 (25%)
Fluorouracil	57 (26%)	51 (23%)
Other chemotherapies‡	64 (29%)	53 (24%)
Other	0	1 (<1%)
Region 1 (Europe, Israel, Australia, United States)	(n=198)	(n=200)
Any treatment	76 (38%)	72 (36%)
Biologic	9 (5%)	5 (3%)
Chemotherapy†	76 (38%)	71 (36%)
Irinotecan	57 (29%)	48 (24%)
Fluorouracil	55 (28%)	47 (24%)
Other chemotherapies‡	60 (30%)	50 (25%)
Other	0	1 (<1%)
Region 2 (Argentina, Brazil, Chile, and Mexico)	(n=23)	(n=21)
Any treatment	7 (30%)	7 (33%)
Biologic	0	1 (5%)
Chemotherapy†	7 (30%)	7 (33%)
Irinotecan	6 (26%)	7 (33%)
Fluorouracil	2 (9%)	4 (19%)
Other chemotherapies‡	4 (17%)	3 (14%)
Other	0	0
Region 3 (Japan, South Korea, Hong Kong, Singapore, and Taiwan)	(n=109)	(n=114)
Any treatment	75 (69%)	75 (66%)
Biologic	14 (13%)	12 (11%)
Chemotherapy	75 (69%)	74 (65%)
Irinotecan	61 (56%)	62 (54%)
Fluorouracil	26 (24%)	23 (20%)
Other chemotherapies‡	61 (56%)	60 (53%)
Other	1 (<1%)	6 (5%)

Data are n (%), unless otherwise indicated. *Europe, Israel, Australia, United States, Argentina, Brazil, Chile, and Mexico. †Patients may be in more than one category; ‡Other chemotherapies include anthracyclines, antineoplastic agents, folic acid analogues, nitrogen mustard analogues, non-steroids, cytotoxic antibiotics, platinum compounds, podophyllotoxin derivatives, pyrimidine analogues, and taxanes.

***Table 8S*: Treatment-emergent adverse events occurring in ≥10% of patients (grades 1-2) in the ramucirumab plus paclitaxel group or any grade ≥3, regardless of causality**

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
Any patients with TEAE	57 (17%)	155 (47%)	73 (22%)	39 (12%)	116 (35%)	128 (39%)	27 (8%)	51 (16%)
Decreased appetite	121 (37%)	10 (3%)	0	0	92 (28%)	13 (4%)	0	0
Nausea	109 (33%)	5 (2%)	1 (<1%)	0	100 (30%)	8 (2%)	0	0
Alopecia	107 (33%)	0	0	0	126 (38%)	1 (<1%)	0	0
Fatigue	107 (33%)	23 (7%)	0	0	93 (28%)	13 (4%)	0	0
Epistaxis	100 (31%)	0	0	0	23 (7%)	0	0	0
Diarrhoea	94 (29%)	12 (4%)	0	0	71 (22%)	4 (1%)	1 (<1%)	0
Abdominal pain	83 (25%)	18 (6%)	0	0	56 (17%)	10 (3%)	1 (<1%)	0
Anaemia	81 (25%)	30 (9%)	0	0	83 (25%)	31 (9%)	3 (<1%)	0
Vomiting	78 (24%)	9 (3%)	1 (<1%)	0	56 (17%)	12 (4%)	0	0
Oedema peripheral	77 (24%)	5 (2%)	0	0	43 (13%)	2 (<1%)	0	0
Constipation	70 (21%)	0	0	0	69 (21%)	2 (<1%)	0	0
Stomatitis	62 (19%)	2 (<1%)	0	0	22 (7%)	2 (<1%)	0	0
Pyrexia	56 (17%)	3 (<1%)	0	0	36 (11%)	1 (<1%)	0	0
Leukopenia	54 (17%)	52 (16%)	5 (2%)	0	47 (14%)	19 (6%)	3 (<1%)	0
Asthenia	51 (16%)	18 (6%)	0	0	39 (12%)	6 (2%)	0	0
Peripheral sensory neuropathy	51 (16%)	6 (2%)	0	0	33 (10%)	3 (<1%)	0	0
Proteinuria	50 (15%)	4 (1%)	0	0	20 (6%)	0	0	0
Neutropenia	45 (14%)	71 (22%)	62 (19%)	0	40 (12%)	51 (16%)	11 (3%)	0
Cough	40 (12%)	0	0	0	25 (8%)	0	0	0
Weight decreased	39 (12%)	6 (2%)	0	0	45 (14%)	4 (1%)	0	0
Thrombocytopenia	38 (12%)	5 (2%)	0	0	14 (4%)	6 (2%)	0	0
Neuropathy peripheral	37 (11%)	10 (3%)	0	0	23 (7%)	7 (2%)	0	0
Back pain	35 (11%)	4 (1%)	0	0	35 (11%)	5 (2%)	0	0
Rash	35 (11%)	0	0	0	26 (8%)	0	0	0
Myalgia	34 (10%)	0	0	0	32 (10%)	1 (<1%)	0	0
Dyspnoea	34 (10%)	8 (2%)	0	0	29 (9%)	2 (<1%)	0	0
Headache	32 (10%)	0	0	0	21 (6%)	1 (<1%)	0	0
Hypertension	32 (10%)	46 (14%)	0	0	8 (2%)	8 (2%)	0	0
Abdominal pain upper	29 (9%)	3 (<1%)	0	0	34 (10%)	1 (<1%)	0	0
Arthralgia	28 (9%)	1 (<1%)	0	0	19 (6%)	1 (<1%)	0	0
Hypoalbuminaemia	27 (8%)	4 (1%)	0	0	11 (3%)	1 (<1%)	0	1 (<1%)
Dry skin	25 (8%)	0	0	0	9 (3%)	1 (<1%)	0	0
Ascites	21 (6%)	11 (3%)	1 (<1%)	0	14 (4%)	13 (4%)	0	0
Aspartate aminotransferase increased	21 (6%)	6 (2%)	0	0	12 (4%)	5 (2%)	0	0
Pruritus	20 (6%)	1 (<1%)	0	0	11 (3%)	0	0	0
Abdominal distension	19 (6%)	2 (<1%)	0	0	16 (5%)	4 (1%)	0	0
Paraesthesia	18 (6%)	6 (2%)	0	0	24 (7%)	1 (<1%)	0	0
Dizziness	18 (6%)	1 (<1%)	0	0	13 (4%)	0	0	0
Urinary tract infection	18 (6%)	1 (<1%)	0	0	10 (3%)	2 (<1%)	0	0
Alanine	16 (5%)	4 (1%)	0	0	15 (5%)	2 (<1%)	1 (<1%)	0

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
aminotransferase increased								
Dysphonia	16 (5%)	1 (<1%)	0	0	9 (3%)	0	0	0
Polyneuropathy	14 (4%)	4 (1%)	0	0	18 (5%)	4 (1%)	0	0
Upper respiratory tract infection	14 (4%)	1 (<1%)	0	0	10 (3%)	0	0	0
Infusion related reaction	13 (4%)	1 (<1%)	0	0	8 (2%)	0	0	0
Blood bilirubin increased	13 (4%)	3 (<1%)	0	0	6 (2%)	5 (2%)	0	0
Dysphagia	12 (4%)	4 (1%)	0	0	12 (4%)	7 (2%)	0	0
Blood creatinine increased	12 (4%)	1 (<1%)	1 (<1%)	0	6 (2%)	2 (<1%)	0	0
Blood alkaline phosphatase increased	11 (3%)	4 (1%)	0	0	9 (3%)	5 (2%)	1 (<1%)	0
Weight increased	11 (3%)	2 (<1%)	0	0	9 (3%)	2 (<1%)	0	0
Hypocalcaemia	11 (3%)	0	0	0	4 (1%)	1 (<1%)	0	0
Hypoaesthesia	9 (3%)	1 (<1%)	0	0	8 (2%)	0	0	0
Hypomagnesaemia	9 (3%)	1 (<1%)	0	0	6 (2%)	0	0	0
Pleural effusion	9 (3%)	3 (<1%)	1 (<1%)	0	2 (<1%)	2 (<1%)	2 (<1%)	0
Hypokalaemia	8 (2%)	6 (2%)	0	0	10 (3%)	5 (2%)	0	0
Mucosal inflammation	8 (2%)	2 (<1%)	0	0	6 (2%)	0	0	0
C-reactive protein increased	8 (2%)	2 (<1%)	0	0	5 (2%)	1 (<1%)	0	0
Hyponatraemia	8 (2%)	8 (2%)	3 (<1%)	0	5 (2%)	3 (<1%)	1 (<1%)	0
Leukocytosis	8 (2%)	1 (<1%)	0	0	4 (1%)	2 (<1%)	0	0
Gamma-glutamyltransferase increased	8 (2%)	1 (<1%)	2 (<1%)	0	3 (<1%)	2 (<1%)	1 (<1%)	0
Depression	7 (2%)	1 (<1%)	0	0	7 (2%)	0	0	0
Pain	7 (2%)	3 (<1%)	0	0	3 (<1%)	1 (<1%)	0	0
General physical health deterioration	6 (2%)	8 (2%)	3 (<1%)	0	7 (2%)	7 (2%)	2 (<1%)	2 (<1%)
Hypotension	6 (2%)	4 (1%)	0	0	6 (2%)	2 (<1%)	0	0
Cancer pain	6 (2%)	1 (<1%)	0	0	5 (2%)	2 (<1%)	0	0
Dehydration	6 (2%)	7 (2%)	1 (<1%)	1 (<1%)	5 (2%)	8 (2%)	0	0
Hyperbilirubinaemia	6 (2%)	1 (<1%)	0	0	5 (2%)	2 (<1%)	0	0
Erythema	6 (2%)	1 (<1%)	0	0	4 (1%)	0	0	0
Haemoptysis	6 (2%)	1 (<1%)	0	0	3 (<1%)	0	0	0
Neurotoxicity	6 (2%)	3 (<1%)	0	0	3 (<1%)	0	0	0
Mouth ulceration	6 (2%)	1 (<1%)	0	0	2 (<1%)	0	0	0
Hyperkalaemia	5 (2%)	5 (2%)	0	0	9 (3%)	2 (<1%)	0	0
Blood lactate dehydrogenase increased	5 (2%)	1 (<1%)	0	0	6 (2%)	1 (<1%)	0	0
Hyperglycaemia	5 (2%)	3 (<1%)	0	0	5 (2%)	5 (2%)	0	0
Oral candidiasis	5 (2%)	1 (<1%)	0	0	3 (<1%)	0	0	0
Blood albumin decreased	5 (2%)	0	0	0	2 (<1%)	1 (<1%)	0	0
Hydronephrosis	5 (2%)	1 (<1%)	0	0	2 (<1%)	1 (<1%)	0	0
Pneumonia	5 (2%)	5 (2%)	0	0	2 (<1%)	4 (1%)	1 (<1%)	0
Malignant neoplasm	5 (2%)	16 (5%)	4 (1%)	27 (8%)	1 (<1%)	24 (7%)	1 (<1%)	34 (10%)

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
progression								
Hyperuricaemia	4 (1%)	1 (<1%)	0	0	5 (2%)	0	0	0
Lethargy	4 (1%)	1 (<1%)	0	0	5 (2%)	0	0	0
Musculoskeletal pain	4 (1%)	1 (<1%)	0	0	5 (2%)	1 (<1%)	0	0
Flank pain	4 (1%)	0	0	0	3 (<1%)	1 (<1%)	0	0
Hypophosphataemia	4 (1%)	4 (1%)	0	0	3 (<1%)	3 (<1%)	1 (<1%)	0
Melaena	4 (1%)	1 (<1%)	1 (<1%)	0	3 (<1%)	0	0	0
Tumour pain	4 (1%)	0	0	0	3 (<1%)	1 (<1%)	0	0
Muscular weakness	4 (1%)	1 (<1%)	0	0	2 (<1%)	3 (<1%)	0	0
Eastern Cooperative Oncology Group performance status worsened	4 (1%)	3 (<1%)	0	0	1 (<1%)	3 (<1%)	0	0
Hypoglycaemia	4 (1%)	0	0	0	1 (<1%)	1 (<1%)	0	0
Lung infection	4 (1%)	1 (<1%)	1 (<1%)	0	1 (<1%)	0	0	0
Vertigo	3 (<1%)	0	0	0	4 (1%)	1 (<1%)	0	0
Conjunctivitis	3 (<1%)	1 (<1%)	0	0	3 (<1%)	0	0	0
Device related infection	3 (<1%)	2 (<1%)	0	0	3 (<1%)	4 (1%)	0	0
Drug hypersensitivity	3 (<1%)	1 (<1%)	0	0	3 (<1%)	0	0	0
Lymphopenia	3 (<1%)	2 (<1%)	0	0	3 (<1%)	0	2 (<1%)	0
Blood pressure increased	3 (<1%)	2 (<1%)	0	0	2 (<1%)	0	0	0
Ejection fraction decreased	3 (<1%)	2 (<1%)	0	0	2 (<1%)	1 (<1%)	0	0
Pneumonitis	3 (<1%)	1 (<1%)	0	0	2 (<1%)	0	0	0
Palmar-plantar erythrodysaesthesia syndrome	3 (<1%)	1 (<1%)	0	0	1 (<1%)	1 (<1%)	0	0
Performance status decreased	3 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Lacrimation increased	2 (<1%)	0	1 (<1%)	0	4 (1%)	0	0	0
Deep vein thrombosis	2 (<1%)	3 (<1%)	0	0	2 (<1%)	1 (<1%)	0	0
Herpes zoster	2 (<1%)	0	0	0	2 (<1%)	1 (<1%)	0	0
Jaundice	2 (<1%)	1 (<1%)	0	0	2 (<1%)	1 (<1%)	0	0
Haematemesis	2 (<1%)	2 (<1%)	0	0	1 (<1%)	0	0	0
Neuralgia	2 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Pneumothorax	2 (<1%)	0	0	0	1 (<1%)	1 (<1%)	0	0
Staphylococcal infection	2 (<1%)	0	0	0	1 (<1%)	1 (<1%)	0	0
Blood urea increased	2 (<1%)	0	1 (<1%)	0	0	0	0	0
Lower respiratory tract infection	2 (<1%)	0	0	0	0	0	0	1 (<1%)
Regurgitation	2 (<1%)	1 (<1%)	0	0	0	0	0	0
Cachexia	1 (<1%)	2 (<1%)	0	0	7 (2%)	4 (1%)	0	1 (<1%)
Chest pain	1 (<1%)	2 (<1%)	0	0	4 (1%)	0	0	0
Bronchopneumonia	1 (<1%)	1 (<1%)	0	0	3 (<1%)	0	0	0
Gastric haemorrhage	1 (<1%)	4 (1%)	0	0	3 (<1%)	2 (<1%)	0	0
Ileus	1 (<1%)	0	1 (<1%)	0	3 (<1%)	1 (<1%)	0	0
Urinary retention	1 (<1%)	2 (<1%)	0	0	3 (<1%)	0	0	0
Abdominal infection	1 (<1%)	1 (<1%)	0	0	2 (<1%)	0	0	0
Atrial fibrillation	1 (<1%)	2 (<1%)	1 (<1%)	0	2 (<1%)	0	0	0

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
Gastrointestinal haemorrhage	1 (<1%)	3 (<1%)	0	1 (<1%)	2 (<1%)	0	1 (<1%)	1 (<1%)
Hypercalcaemia	1 (<1%)	1 (<1%)	0	0	2 (<1%)	0	0	0
Infection	1 (<1%)	2 (<1%)	0	0	2 (<1%)	0	0	0
Intestinal obstruction	1 (<1%)	3 (<1%)	2 (<1%)	0	2 (<1%)	3 (<1%)	0	0
Tremor	1 (<1%)	1 (<1%)	0	0	2 (<1%)	0	0	0
Oesophagitis	1 (<1%)	3 (<1%)	0	0	1 (<1%)	0	0	0
Skin exfoliation	1 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Syncope	1 (<1%)	2 (<1%)	0	0	1 (<1%)	1 (<1%)	0	0
Wound infection	1 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Angina pectoris	1 (<1%)	0	0	0	0	1 (<1%)	0	0
Cardiac failure	1 (<1%)	0	0	0	0	0	0	1 (<1%)
Cholecystitis	1 (<1%)	2 (<1%)	0	0	0	0	0	0
Depressed level of consciousness	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Hearing impaired	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Hypertransaminasaemia	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Neutrophil count decreased	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Obstruction gastric	1 (<1%)	1 (<1%)	0	0	0	1 (<1%)	0	0
Oesophageal food impaction	1 (<1%)	0	0	0	0	1 (<1%)	0	0
Onycholysis	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Periodontitus	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Peripheral motor neuropathy	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Peritonitis	1 (<1%)	5 (2%)	1 (<1%)	0	0	0	0	0
Upper gastrointestinal haemorrhage	1 (<1%)	0	0	0	0	1 (<1%)	0	0
Spinal pain	0	0	0	0	3 (<1%)	1 (<1%)	0	0
Interstitial lung disease	0	0	0	0	2 (<1%)	2 (<1%)	0	0
Renal failure acute	0	1 (<1%)	0	0	2 (<1%)	0	1 (<1%)	2 (<1%)
Acute respiratory distress syndrome	0	1 (<1%)	0	0	1 (<1%)	0	0	0
Arthritis	0	1 (<1%)	0	0	1 (<1%)	0	0	0
Blood pressure decreased	0	1 (<1%)	0	0	1 (<1%)	0	0	0
Convulsion	0	1 (<1%)	0	0	1 (<1%)	1 (<1%)	0	0
Disease progression	0	0	0	0	1 (<1%)	1 (<1%)	0	0
Feeling abnormal	0	0	0	0	1 (<1%)	1 (<1%)	0	0
Hepatic enzyme increased	0	3 (<1%)	0	0	1 (<1%)	0	0	0
Hypophagia	0	1 (<1%)	0	0	1 (<1%)	1 (<1%)	0	0
Jaundice cholestatic	0	0	0	0	1 (<1%)	1 (<1%)	0	0
Liver function test abnormal	0	1 (<1%)	1 (<1%)	0	1 (<1%)	0	0	0
Procedural pain	0	0	0	0	1 (<1%)	2 (<1%)	0	0
Pulmonary embolism	0	3 (<1%)	0	1 (<1%)	1 (<1%)	7 (2%)	1 (<1%)	2 (<1%)
Renal failure	0	2 (<1%)	1 (<1%)	0	1 (<1%)	0	0	0
Small intestinal obstruction	0	3 (<1%)	0	0	1 (<1%)	3 (<1%)	0	0
White blood cell count	0	1 (<1%)	0	0	1 (<1%)	0	0	0

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
decreased								
Acute coronary syndrome	0	0	0	0	0	1 (<1%)	0	0
Acute prerenal failure	0	0	0	0	0	1 (<1%)	0	0
Akathisia	0	0	0	0	0	1 (<1%)	0	0
Amylase increased	0	0	0	0	0	1 (<1%)	0	0
Anal abscess	0	1 (<1%)	0	0	0	0	0	0
Anticoagulation drug level above therapeutic	0	1 (<1%)	0	0	0	0	0	0
Aphagia	0	1 (<1%)	0	0	0	0	0	0
Aplastic anaemia	0	1 (<1%)	0	0	0	0	0	0
Aspiration	0	0	0	0	0	0	0	1 (<1%)
Atrial thrombosis	0	0	0	0	0	1 (<1%)	0	0
Bile duct obstruction	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Biliary sepsis	0	0	1 (<1%)	0	0	0	0	0
Biliary tract infection	0	0	0	0	0	1 (<1%)	0	0
Brain cancer metastatic	0	0	1 (<1%)	0	0	0	0	0
Catheter site infection	0	0	0	0	0	1 (<1%)	0	0
Cellulitis	0	0	0	0	0	1 (<1%)	0	0
Cerebral haemorrhage	0	0	0	0	0	0	0	1 (<1%)
Cerebrovascular accident	0	0	1 (<1%)	0	0	0	0	0
Cholangitis	0	0	0	0	0	1 (<1%)	1 (<1%)	0
Cholecystitis acute	0	1 (<1%)	0	0	0	0	0	0
Clavicle fracture	0	1 (<1%)	0	0	0	0	0	0
Cognitive disorder	0	1 (<1%)	0	0	0	0	0	0
Coma	0	0	1 (<1%)	0	0	0	0	0
Death	0	0	0	1 (<1%)	0	0	0	0
Device dislocation	0	0	0	0	0	0	1 (<1%)	0
Diabetes insipidus	0	0	0	0	0	1 (<1%)	0	0
Diabetes mellitus	0	1 (<1%)	0	0	0	0	0	0
Diarrhoea infectious	0	0	0	0	0	1 (<1%)	0	0
Disseminated intravascular coagulation	0	2 (<1%)	0	0	0	0	1 (<1%)	0
Diverticular perforation	0	0	1 (<1%)	0	0	0	0	0
Drowning	0	0	0	0	0	0	0	1 (<1%)
Endocarditis	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Extrasystoles	0	0	0	0	0	1 (<1%)	0	0
Extremity contracture	0	1 (<1%)	0	0	0	0	0	0
Febrile neutropenia	0	8 (2%)	2 (<1%)	0	0	4 (1%)	3 (<1%)	1 (<1%)
Gallbladder obstruction	0	0	0	0	0	1 (<1%)	0	0
Gastrointestinal perforation	0	1 (<1%)	0	1 (<1%)	0	0	0	0
Generalised oedema	0	0	0	0	0	1 (<1%)	0	0
Hydrocephalus	0	0	0	0	0	0	1 (<1%)	0
Hypokinesia	0	0	0	0	0	1 (<1%)	0	0
Hypovolaemic shock	0	0	0	0	0	0	0	1 (<1%)
Hypoxia	0	1 (<1%)	0	0	0	0	0	0

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
Inappropriate antidiuretic hormone secretion	0	1 (<1%)	0	0	0	0	0	0
Incisional hernia	0	1 (<1%)	0	0	0	0	0	0
Injection site injury	0	1 (<1%)	0	0	0	0	0	0
Intestinal ischaemia	0	0	0	0	0	1 (<1%)	0	0
Intestinal perforation	0	0	1 (<1%)	0	0	0	0	0
Intracranial pressure increased	0	0	0	0	0	0	0	1 (<1%)
Ischaemic stroke	0	0	1 (<1%)	0	0	0	0	0
Laryngitis	0	0	0	0	0	1 (<1%)	0	0
Lobar pneumonia	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Localised infection	0	1 (<1%)	0	0	0	0	0	0
Lung infiltration	0	1 (<1%)	0	0	0	0	0	0
Malabsorption	0	0	0	1 (<1%)	0	0	0	0
Malignant ascites	0	0	0	0	0	2 (<1%)	0	0
Malignant pleural effusion	0	0	0	0	0	1 (<1%)	0	0
Malnutrition	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Metastases to central nervous system	0	0	0	0	0	1 (<1%)	0	0
Metastases to liver	0	0	0	0	0	1 (<1%)	0	0
Metastases to meninges	0	1 (<1%)	0	0	0	2 (<1%)	0	0
Multi-organ failure	0	0	0	0	0	2 (<1%)	0	0
Myocardial infarction	0	1 (<1%)	0	0	0	0	0	1 (<1%)
Neutropenic sepsis	0	0	0	0	0	0	1 (<1%)	0
Obstructive uropathy	0	1 (<1%)	0	0	0	0	0	0
Oesophageal haemorrhage	0	1 (<1%)	0	0	0	0	0	0
Oesophageal spasm	0	1 (<1%)	0	0	0	0	0	0
Oesophageal stenosis	0	0	0	0	0	1 (<1%)	0	0
Oliguria	0	0	0	0	0	1 (<1%)	0	0
Pain of skin	0	1 (<1%)	0	0	0	0	0	0
Pancreatitis acute	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Pathological fracture	0	0	0	0	0	1 (<1%)	0	0
Perirectal abscess	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Peritoneal haemorrhage	0	0	0	0	0	1 (<1%)	0	0
Peritonitis bacterial	0	2 (<1%)	1 (<1%)	0	0	2 (<1%)	0	0
Pneumatosis intestinalis	0	1 (<1%)	0	0	0	0	0	0
Portal vein thrombosis	0	2 (<1%)	0	0	0	0	0	0
Procedural hypertension	0	1 (<1%)	0	0	0	1 (<1%)	0	0
Pseudomonas infection	0	1 (<1%)	0	0	0	0	0	0
Pulmonary sepsis	0	0	0	0	0	0	1 (<1%)	0
Pulmonary thrombosis	0	1 (<1%)	0	0	0	0	0	0
Pyloric stenosis	0	0	0	0	0	1 (<1%)	0	0
Respiratory failure	0	0	0	1 (<1%)	0	1 (<1%)	0	0
Sepsis	0	0	3 (<1%)	2 (<1%)	0	0	2 (<1%)	0
Septic shock	0	0	0	3 (<1%)	0	0	0	1 (<1%)
Small intestinal stenosis	0	0	0	0	0	1 (<1%)	0	0
Spinal cord compression	0	0	0	0	0	1 (<1%)	0	0

Preferred Term (MedDRA v16.0)	Ramucirumab plus Paclitaxel (n=327)				Placebo plus Paclitaxel (n=329)			
	Grades 1-2	Grade 3	Grade 4	Grade 5	Grades 1-2	Grade 3	Grade 4	Grade 5
Splenic rupture	0	1 (<1%)	0	0	0	0	0	0
Staphylococcal sepsis	0	1 (<1%)	0	0	0	0	0	0
Stress cardiomyopathy	0	0	0	0	0	1 (<1%)	0	0
Systolic hypertension	0	1 (<1%)	0	0	0	0	0	0
Thoracic vertebral fracture	0	1 (<1%)	0	0	0	0	0	0
Urinary tract infection bacterial	0	1 (<1%)	0	0	0	0	0	0
Vena cava thrombosis	0	0	0	0	0	2 (<1%)	0	0

Data are n (%) unless otherwise stated. MedDRA= Medical Dictionary for Regulatory Activities; TEAE=treatment-emergent adverse events.

List of Investigators

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Brazil	123	Dr Renata Fernandino Garcia	Belo Horizonte	Minas Gerais	Oxion Medicina Oncológica [Clinical Research Centre]
Brazil	126	Dr Marcelo Tanaka	São Paulo	São Paulo	Hospital Santa Cruz [Clinical Research Centre]
Brazil	127	Dr Jose Alberto Lopes Nogueira	Salvador	Bahia	Oncovida - Centro de Oncologia da Bahia [Centro Médico Iguatemi]
Brazil	129	Dr Rodrigo Cunha Guimarães	Belo Horizonte	Minas Gerais	Hospital das Clinicas da UFMG [Hematology and Oncology]
Brazil	130	Dr Roberto Odebrecht Rocha	São Paulo	São Paulo	Hospital Santa Marcelina [Centro de Pesquisa Clínica]
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Brazil	132	Dr Fernando Meton Camara Vieira	Rio de Janeiro	Rio de Janeiro	Inca- Instituto Nacional do Câncer
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Brazil	135	Dr Leandro Brust	Rio Grande do Sul	Rio Grande do Sul	Hospital Bruno Born [Oncology’
Brazil	136	Dr Luis Alberto Schlittler	Passo Fundo	Rio Grande do Sul	Hospital da Cidade de Passo Fundo [Centro de Pesquisa em Oncologia]
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Brazil	138	Dr Fabio Andre Franke	Ijuí	Rio Grande do Sul	Hospital de Caridade de Ijuí [Clinical Research Centre]
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Australia	204	Prof Dusan Kotasek	Kurralta Park	South Australia	Ashford Cancer Centre Research [Haematology]
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Australia	207	Dr Marco Matos	Southport	Queensland	Gold Coast University Hospital [Oncology]
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Belgium	242	Dr Alain Hendlisz	Bruxelles		Jules Bordet Instituut
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Romania	447	Dr Constantin Volovat	Lasi	Lasi	Centrul de Oncologie Euroclinic [Oncology]
Romania	448	Dr Dumitru Filip	Baia-Mare	Maramureş	Spitalul Judetean de Urgenta "Dr. Constantin Opris" [Oncologie Medicala]
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Russian Federation	461	Prof Oleg Nikolaevich Lipatov	Ufa	Bashkortostan, Respublika	Republican Clinical Oncology Dispensary [Oncology]
Russian Federation	462	Prof Sergey Tjulandin	Moscow		Russian Oncological Research Center n.a. N.N. Blokhin RAMS [Pharmacology and Chemotherapy]
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Spain	484	Dr José Alés	Avila		Complejo Asistencial de Avila-Hospital Nuestra Señora de Sonsoles [Oncología]
Spain	485	Dr David Vicente	Sevilla		Hospital Universitario Virgen de la Macarena Servicio de Oncología
Spain	486	Ms. María Luisa Limón Mirón	Sevilla	Andalucía	Hospital Universitario Virgen del Rocío Servicio de Oncología
Spain	488	Dr Antonio Sánchez Ruiz	Madrid	Madrid	Hospital Universitario Puerta de Hierro-Majadahonda Servicio de Oncología
Spain	489	Dr. José A. López Martín	Madrid		Hospital Universitario 12 de Octubre Servicio de Oncología Medica
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United Kingdom	504	Prof David Cunningham	London		Royal Marsden Hospital

United Kingdom	505	Prof David Ferry	Wolverhampton		New Cross Hospital [Oncology]
Estonia	520	Dr Kristiina Ojamaa	Tallinn	Harjumaa	East Tallinn Central Hospital [Internal Medicine]
Estonia	521	Dr Tiit Suuroja	Tallinn		North Estonia Medical Centre
Hong Kong	730	Prof Kent Man 建民 Chu 朱	Hong Kong		Queen Mary Hospital [Surgery]
Hong Kong	731	Dr Sing Hung 勝雄 Lo 魯	Hong Kong		Tuen Mun Hospital [Clinical Oncology]
Japan	801	Prof Dr Yoshito 嘉人 Komatsu 小松	Sapporo	Hokkaidô	Hokkaido University Hospital [Gastroenterology]
Japan	802	Dr Seigo斉悟Yukisawa 行澤	Utsunomiya	Tochigi	Tochigi Cancer Center [Medical Oncology]
Japan	803	Dr Hirofumi 博文 Fujii 藤井	Shimotsuke	Tochigi	Jichi Medical University Hospital [Medical Oncology]
Japan	804	Dr Toshihiko 俊彦 Doi 土井	Kashiwa	Tiba	National Cancer Center Hospital East [Gastrointestinal Oncology]
Japan	805	Dr Kensei 研成 Yamaguchi 山口	Kitaadachi-gun	Saitama	Saitama Cancer Center [Gastroenterology]
Japan	806	Dr Yoshihiko 芳彦 Segawa 畝川	Hidaka	Saitama	Saitama Medical University International Medical Center [Medical Oncology]
Japan	807	Dr Shuichi 秀一 Hironaka 廣中	Chiba	Tiba	Chiba Cancer Center [Clinical Trial Promotion]
Japan	808	Dr Yasuhiro 安博 Shimada 島田	Chuo-ku	Tôkyô	National Cancer Center Hospital [Gastrointestinal Medical Oncology]
Japan	809	Dr Yasushi 泰司 Omuro 小室	Bunkyo-ku	Tôkyô	Tokyo Metropolitan Cancer and Infectious Diseases Center Komagome HP [Chemotherapy]
Japan	810	Dr Hirofumi 博史 Yasui 安井	Sunto-gun	Sizuoka	Shizuoka Cancer Center [Gastrointestinal Oncology]
Japan	811	Dr Kei 圭 Muro 室	Nagoya	Aiti	Aichi Cancer Center Hospital [Clinical Oncology]
Japan	812	Dr Naotoshi 直俊 Sugimoto 杉本	Osaka	Ôsaka	Osaka Medical Center for Cancer and Cardiovascular Diseases [Clinical Oncology]
Japan	813	Dr Fumio 文夫 Nagashima 長島	Mitaka	Tôkyô	Kyorin University Hospital [Medical Oncology]
Japan	814	Dr Masahiro 昌弘 Gotoh 後藤	Takatsuki	Ôsaka	Osaka Medical College Hospital [Cancer Chemotherapy Center]
Japan	815	Dr Takao 孝雄Tamura 田村	Osaka-Sayama	Ôsaka	Kinki University Hospital [Medical Oncology]
Japan	816	Dr Motohiro 素宏Hirao 平尾	Osaka	Ôsaka	Osaka National Hospital [Digestive Organs]
Japan	817	Dr Tomohiro 智裕 Nishina 仁科	Matsuyama	Ehime	Shikoku Cancer Center [Gastroenterology]
Japan	818	Dr Sojiro 荘二郎 Morita 森田	Kochi	Kôti	Kochi Health Sciences Center [Medical Oncology]
Japan	819	Dr Taito 泰斗 Esaki 江崎	Fukuoka	Fukuoka	National Kyushu Cancer Center [Gastrointestinal Medical Oncology]
Japan	820	Dr Yoshinori 詳典 Hirashima 平島	Yufu	Ôita	Oita University Hospital [Medical Oncology]
Korea, Republic Of	830	Prof Sang Cheul 상철 Oh 오	Seoul	Seoul Teugbyeolsi	Korea University Guro Hospital [Hematology-Oncology]
Korea, Republic Of	832	Prof Hoon-Kyo Kim	Suwon	Gyeonggido	Catholic University of Korea, St. Vincent's Hospital [Oncology]

Korea, Republic Of	833	Dr Keun Wook 근욱 Lee 이	Seoungnam	Gyeonggido	Seoul National University Bundang Hospital [Hematology/Oncology]
Korea, Republic Of	834	Prof Tae You태유 Kim 김	Seoul		Seoul National University Hospital [Hematology-Oncology]
Korea, Republic Of	835	Dr Dong Bok Shin	Incheon		Gachon University Gil Hospital [Hematology-Oncology]
Korea, Republic Of	836	Dr SeokYun Kang	Suwon	Gyeonggido	Ajou University Hospital [Hematology/Oncology]
Singapore	860	Dr Peter Cher Siang Ang	Singapore		OncoCare Cancer Centre Gleneagles Medical Centre
Singapore	861	Dr Akhil Chopra	Singapore	Central Singapore	Johns Hopkins Singapore International Medical Center [Oncology]
Taiwan	880	Dr Chia-Jui 家瑞 Yen 顏	Tainan		National Cheng Kung University Hospital [Oncology]
Taiwan	881	Prof Yee 毅 Chao 趙	Taipei		Veterans General Hospital-Taipei [Cancer Tumor Center]
Taiwan	882	Dr Chang-Fang 昌芳 Chiu 邱	Taichung		China Medical University Hospital [Oncology and Hematology]
Taiwan	883	Dr Yen-Yang 彥仰 Chen 陳	Kaohsiung		Chang Gung Medical Foundation - Kaohsiung Chang Gung Memoria [Hemato-Oncology]
Taiwan	884	Dr Cheng-Shyong 正雄 Chang 張	Changhua	Taichung	Changhua Christian Hospital [Oncology]
Taiwan	885	Dr Chien-Liang 建良 Lin 林	Liou Ying Township	Tainan	Chi-Mei Medical Center [Hematology and Oncology]